



Data Driven Education

Discussion Forum Subteam

Presentation 2 - 10/22/2025

Meet the Team



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Introduction

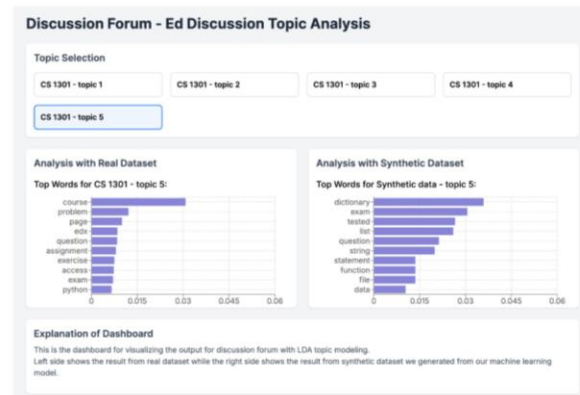
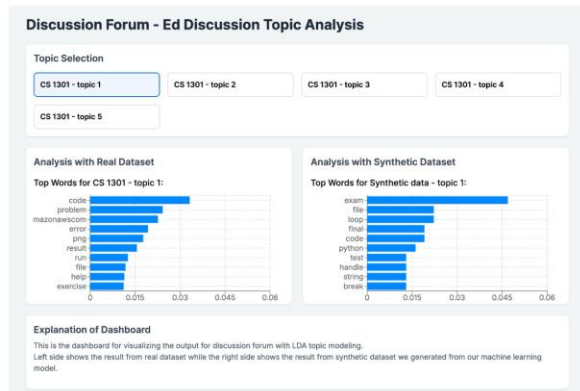
- Online Discussion platform is crucial in modern education and digital studies.
- **Ed Discussion and EdX:** widely used online discussion forum
 - EdX offers Massive Open Online Courses (MOOCs)
 - enables students and instructors to interact
 - discuss course materials asynchronously
- Previous
 - What are some unique characteristics of Online Discussion Forums?
 - What to do to utilize Online Discussion platforms better?
 - Improve the whole education experience
- How to predict student's success from their engagements?

Research Questions

- How to combine sentiment analysis and cognitive presences to provide insights into student engagement and attitudes toward the course?
- How to use machine learning techniques to predict students' success in the course based on their discussion engagements?
 - ~~○ What features should we use for Time Series Analysis?~~
 - How to improve Ridge Regression model performance?
- What is the best way to present discussion analytics to instructors in an actionable and user-friendly dashboard?

Goals for the Semester

- Enhance analysis of online discussion forum data
 - Data-driven insights, ML models
- Explore sentiment analysis with cognitive presences utilizing BERT
- Identify key factors and analyze students' grades in relation to online engagement with Ridge Regression
- Present results with visualization, improve our previous dashboard



Literature Review

Literature Review:

Goals:

- Identify preprocessing and tokenization strategies that improve performance and interpretability within our project's sentiment pipeline.
- Examine engagement metrics and labeling strategies from past studies to guide how we categorize sentiment and measure student participation in our project.
- Investigate how the application of a Ridge regression-based model can inform optimization of teaching environments and analytics integration in educational settings.

Body of Knowledge:

- "Sentiment Analysis with BERT: A Comprehensive Guide" (Alex Rodrigues, 2024)
 - Provides an in-depth walkthrough of BERT's architecture, tokenizer, and fine-tuning process for sentiment analysis.
- "An Investigation on the Application of Ridge Regression Model in the Optimization of Virtual Practice Teaching Innovation Path of Civics and Politics Courses in Colleges and Universities." (Guizhen He & Jianing Zhang, 2024)
 - Uses a Ridge regression model to analyze how virtual teaching environments impact the learning effect in civics/politics courses in higher education.

Literature Takeaways:

“Sentiment Analysis with BERT: A Comprehensive Guide.” (Alex Rodrigues, 2024)

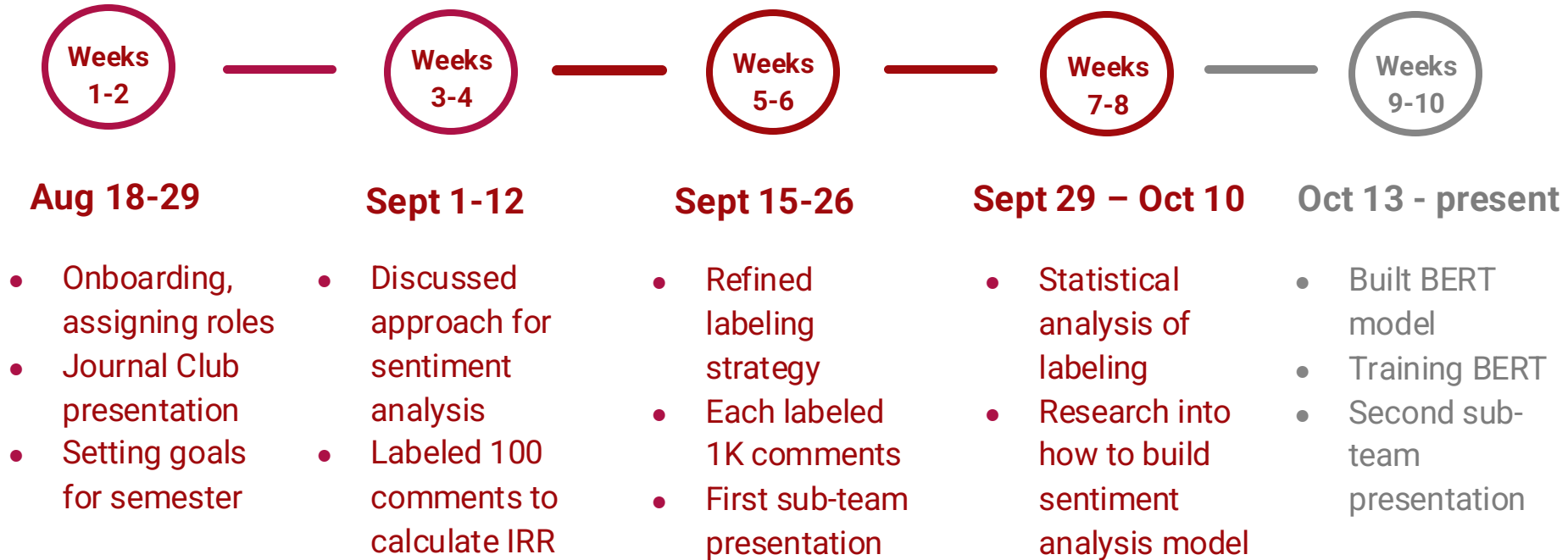
- Demonstrates how transformer-based models capture contextual meaning beyond traditional word embeddings.
- Emphasizes the importance of preprocessing, tokenization, and evaluation metrics like F1-score and accuracy.
- Suggests that contextual embeddings improve sentiment classification in complex text environments.

“An Investigation on the Application of Ridge Regression Model in the Optimization of Virtual Practice Teaching Innovation Path of Civics and Politics Courses in Colleges and Universities.” (Guizhen He & Jianing Zhang, 2024)

- Applies Ridge regression to evaluate key factors influencing virtual teaching effectiveness.
- Demonstrates the value of regularization for reducing overfitting in educational data models.
- Provides a framework linking data-driven modeling to pedagogical innovation in online learning contexts.

Data Science

Goals & Timeline for the Semester



Current Progress

- Manually labelled first thousand edX dataset comments as negative (1), neutral (2), or positive (3)
- Analyzed data to determine majority and average ratings per comment
- Expensive short-term cost --> cheap long-term gain

Clara	Unnamed: Ethan	Unnamed: Saloni	Unnamed: Unnamed: body	majority	avg_vote
1	1	1	Enter your	1	1
2	2	2	SmartBool	2	2
1	1	1	What is th	1	1
1	1	1	Passed wit	1	1
2	1	2	I had the s	2	2
2	2	2	ah, I see. /	2	2
2	2	2	I knew in r	2	2
1	1	1	I upgradec	1	1
3	3	3	I'm interes	3	3
2	2	2	Hi, no I ha	2	2
2	3	3	I'd be inter	3	3
1	1	2	The questi	1	1
1	1	1	lol... I didn	1	1
2	2	2	No, it's no	2	2
2	2	2	Have you l	2	2
3	2	3	Hey PERSC	3	3
2	3	2	Yes, I am	2	2
2	3	3	I am intere	3	3
3	3	3	Ernest, I ai	3	3
3	3	3	Yeah, I am	3	3
2	2	2	Hi, will you	2	2
3	3	3	Hi PERSON	3	3
2	2	2	I'm on a Pl	2	2
3	3	3	Nice to se	3	3
2	2	2	I have a iss	2	2
2	2	2	Ok, so I fig	2	2
2	2	1	can you gi	2	2

Current Progress, cont.

- Create BERT model using first 1k data points + correlated ratings
- 70/15/15 training/validation/testing split to ensure accuracy
- Eval Loss: measures how well the model's predictions match the actual labels

```
from transformers import TrainingArguments

training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=16, # Increased if GPU memory allows
    per_device_eval_batch_size=16,
    warmup_steps=100, # Increased for better warmup
    weight_decay=0.01,
    logging_dir='./logs',
    # max_steps=10, # Comment out or remove to allow full training
    save_steps=100,
    logging_steps=50,
    eval_strategy="epoch", # Add evaluation after each epoch
    save_strategy="epoch",
    load_best_model_at_end=True,
)

trainer = Trainer(
    model=model, # The pre-trained model that you want to fine-tune or train
    args=training_args, # The training arguments that specify the configurations for the training process
    train_dataset=train_dataset, # The dataset used for training the model
    eval_dataset=val_dataset # The dataset used for evaluating the model during training
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/
warnings.warn(warn_msg)
```

[30/30 01:05]

Evaluation Results:

- Loss: 0.9741
- Runtime: 67.99 seconds
- Samples per Second: 2.21
- Steps per Second: 0.44
- Epoch: 0.0714

```
warnings.warn(warn_msg)
```

[13]

Epoch	Training Loss	Validation Loss
1	No log	0.674487
2	0.938600	0.571436
3	0.620800	0.550698

Example prediction

```
example_text = "Wow, I am so impressed!"
predicted_sentiment = predict_sentiment(example_text)
print(f"Predicted Sentiment: {predicted_sentiment}")
```

Predicted Sentiment: positive

Next Steps



Mid-End
Oct

Oct 20 - 29

- Finish training & fine-tuning BERT
- Run BERT on entire dataset to determine student sentiment



Early -
Mid Nov

Nov 3 - 14

- Analyze dataset to determine student levels of cognitive presence
- Explore relationships b/w sentiment analysis and cognitive presence



Mid Nov
- Dec

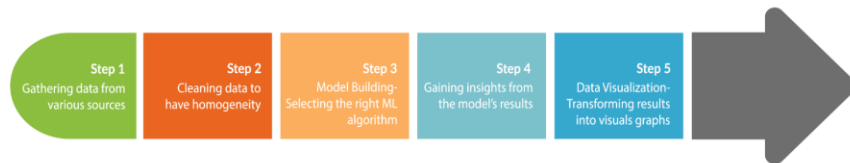
Nov 14 - Dec 2

- Using insights, revamp dashboard from last semesters
- Collaborate with ML team
- Third sub-team presentation

Machine Learning

Goals for the Semester

- Using data from CS1301 2017 & 2018
 - Join grades, forum post, and forum thread datasets with queries
 - Clean null/blank information
- Process data to create a bi-modal student behaviors model
 - Our target variable is student grades and we want to identify which forum behaviors predict academic success
 - We want to study 2 topics:
 - Text embeddings (semantics, etc.)
 - Time-series (frequency, timing in sem, etc.)
- Once identifying which behaviors are key after fusing our bi-modal model, we can create visualizations of those traits and their correlation to grades



Joining Data Tables

How did we compile our key data points?

- Needed to find users that had grade data
 - Pinpointed the classes with that requirement
- Joined on the "user_id" field using Pandas
 - Assured that each column had a grade

```
threads = pd.read_csv(THREADS_CSV, encoding="utf-8", on_bad_lines="skip")
comments = pd.read_csv(COMMENTS_CSV, encoding="utf-8", on_bad_lines="skip")
grades = pd.read_csv(GRADES_CSV, encoding="latin1", on_bad_lines="skip")

for df in [threads, comments]:
    if "created_at" in df.columns:
        df["created_at"] = pd.to_datetime(df["created_at"], errors="coerce")
for df in [threads, comments, grades]:
    if "user_id" in df.columns:
        df["user_id"] = pd.to_numeric(df["user_id"], errors="coerce").astype("Int64")

# Ensure grade column exists
if "percent_grade" not in grades.columns:
    raise ValueError("The grades CSV must contain a 'percent_grade' column (0-1).")
```

id	user_id	course_id	thread_id	title	body	created_at	upvotes	downvotes	comment_type	percent_grade
\x3538613	2771	course-v1:	\x3538393766396134	Hi all, my n	#####		0	0	0 comment	
\x3538633	5710	course-v1:	\x3538633039383339	I am also g	#####		0	0	0 comment	0.85
\x3538646	5710	course-v1:	\x3538643332393134	Firstly, rath	#####		0	0	0 comment	0.85
\x3538646	5710	course-v1:	\x3538646 4.3.4	sort/ My code w	#####		0	0	2 thread	0.85
\x3538656	5710	course-v1:	\x3538656531373464	4.4.4. Exer	#####		0	0	0 comment	0.85
\x3538656	5710	course-v1:	\x3538656535383130	Progress C	#####		0	0	0 comment	0.85
\x3538663	5710	course-v1:	\x3538656531373464	Yes thanks	#####		0	0	0 comment	0.85
\x3539303	5710	course-v1:	\x3539303035333336	I believe th	#####		0	0	0 comment	0.85

Findings from Preliminary EDA:

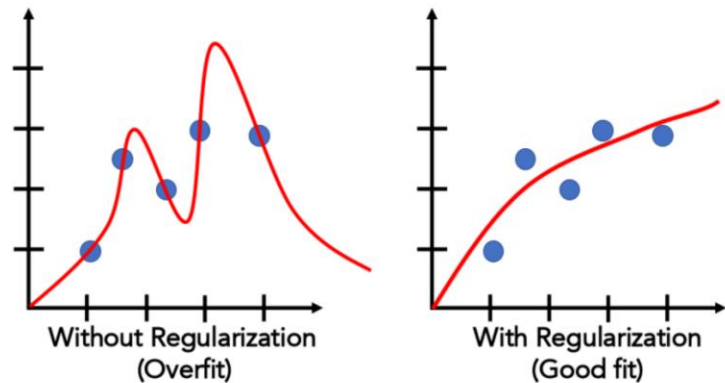
Our target behavior is student grade and we want to pinpoint important student behaviors leading to success:

- It is key to compare between students grades in the discussion forum instead of between classes because difficulty can vary
 - (e.g., CS 1301x average grade 89% vs. CS 1301xl average grade 45%)
- Looking at the top 5 factors in univariate correlation analysis with grades:
 - Active Weeks (consistent posting): +22%
 - Topic Alignment (distance to centroid calculation, on a post/comment level): -21%
 - Weekly Posting Variance (inconsistent posting): -16%
 - Discussion Focus (entropy, on a overall student level): -14%
 - Thread Length: +14%

Modeling Student Grades

How a Ridge Regression model works:

- Uses multiple behavioral features (posting frequency, topic alignment, text length, etc.) together to predict student grades
- Adds a regularization penalty that shrinks less important coefficients, preventing overfitting

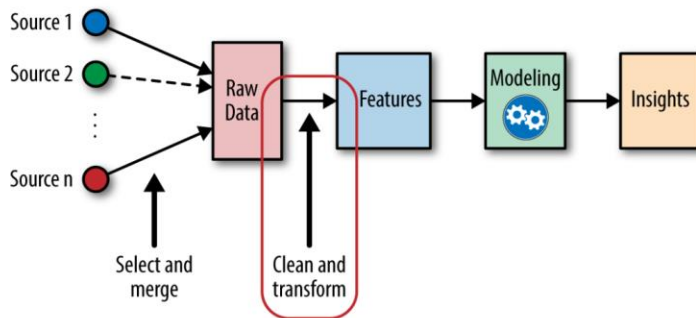


Our multivariate findings with Ridge Regression:

- Model explained about 8.6% of the variance ($R^2 = 0.086$) in student grades, not great but a start
- The top predictors were:
 - Topic Alignment – students writing off-topic from average discussion posts tended to have lower grades (e.g., how can I download X?, is my Python version up to date?, etc.)
 - Active Weeks – consistent participation was strong
 - Discussion Focus – more scattered discussion topics tended to perform worse

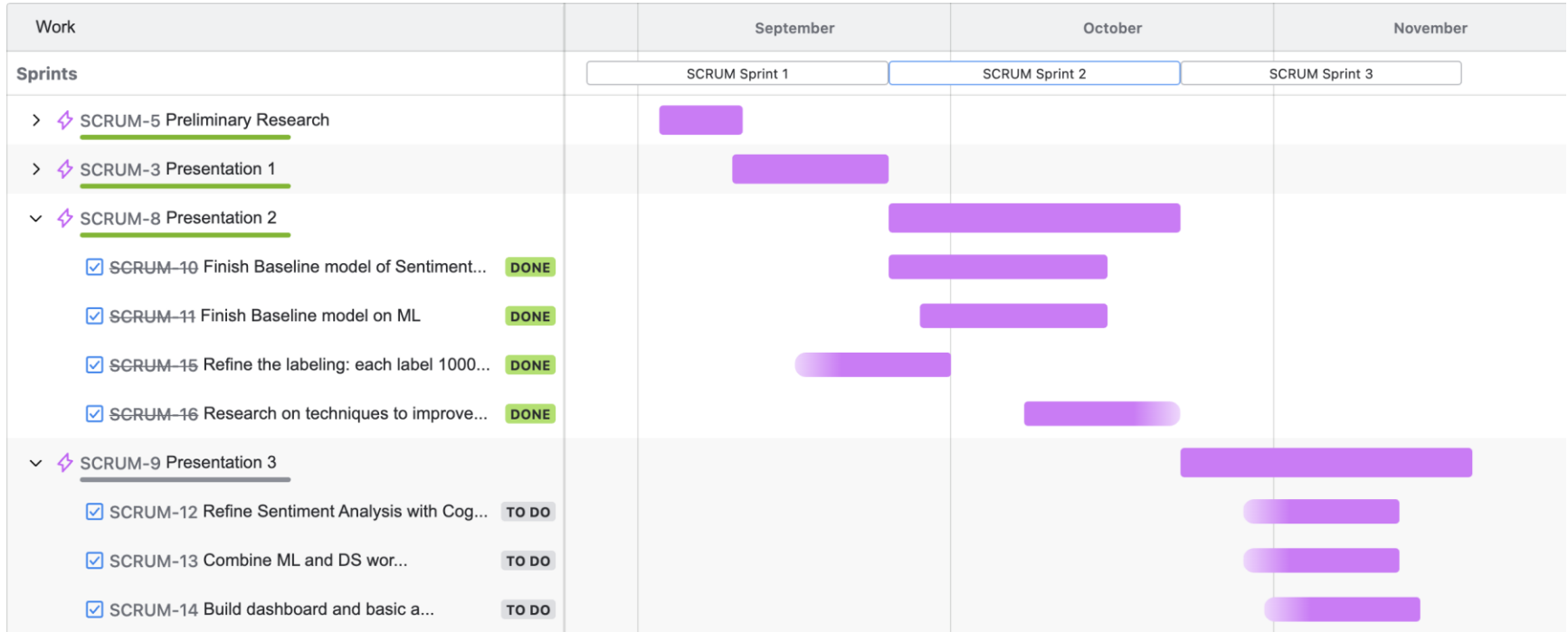
Next Steps:

- Since there is a solid model with key features to dive into, it is important to focus on feature engineering to bring the R^2 up and remove noise
- Visualizing the effects of certain features on grades will also add color to the student behavior trends
- Pushing forward on our dashboard with insights from the Ridge Regression model would be helpful for future groups



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Project Timeline



Jira - For task tracking and management

Conclusion

- Improve course design and refine teaching strategies
- Provide inclusive learning environments
- Personalize learning for foreseeable weakness
- Promote student's engagements and grades
- Ensure a more responsive and adaptive online learning experience.
- Progress: Finished baseline model on both data science and machine learning
- Next step: connect the models and build visualizations for explainability

Citations

1. Rodrigues, Alex. “Sentiment Analysis with BERT: A Comprehensive Guide.” *Medium*, 4 June 2024, medium.com/@alexrodriguesj/sentiment-analysis-with-bert-a-comprehensive-guide-6d4d091eb6bb#6318.
2. He, Guizhen, and Jianing Zhang. “An Investigation on the Application of Ridge Regression Model in the Optimization of Virtual Practice Teaching Innovation Path of Civics and Politics Courses in Colleges and Universities.” *Applied Mathematics and Nonlinear Sciences*, vol. 9, no. 1, 2024, pp. 1-15. DOI:10.2478/amns-2024-1638.

Thanks for listening!

Q&A